



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

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Abstract

This project focuses on the design and implementation of an autopilot system for the longitudinal dynamics of an aircraft, with the goal of achieving control and stability over vertical speed and pitch rate. The primary objectives include developing a vertical speed guidance loop, integrated with an inner pitch rate control loop, and an auto-throttle system to regulate airspeed. Through theoretical analysis and simulation, the system's performance was evaluated based on key criteria, including response time, stability, and robustness to external disturbances. A 6DOF nonlinear simulation was used to validate the control laws, demonstrating the autopilot's ability to track vertical speed and airspeed commands with minimal overshoot and fast settling times. The system's response to gust inputs was also assessed to evaluate its disturbance rejection capabilities. The results confirm that the proposed longitudinal autopilot design meets the specified performance requirements, providing a reliable solution for maintaining desired flight paths and stability in the longitudinal plane.

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1 Introduction

Automatic Flight Control Systems (AFCS) are the key system used in modern aviation, providing stability and control to aircraft, while reducing pilot workload and increasing overall flight safety. AFCS typically incorporate multi-loop control architectures to manage the various degrees of freedom involved in the lateral and longitudinal movements of the aircraft's dynamics. The longitudinal control of an aircraft, which focuses on controlling vertical speed and pitch angle, is used for flight phases such as climb, descent, and level flight.

The foundation of AFCS design relies on control theory principles. According to [1], AFCS are structured around feedback control loops that use sensor data to continuously monitor and adjust aircraft dynamics to ensure the aircraft's stability during flight command inputs. These control loops are constructed to specific flight parameters, such as pitch rate and vertical speed, by typically using Proportional-Integral-Derivative (PID) control, to achieve the desired response. In recent years, advancements in control theory and computational power have further advanced AFCS that can handle multi-input, multi-output (MIMO) control problems, and aids in managing the interactions between control surfaces and actuators.

This study aims to design and implement an autopilot system specifically for the longitudinal dynamics of an aircraft. The design process includes the development of a vertical speed guidance loop, integrated with an inner pitch rate controller, and an auto-throttle control loop for airspeed management. The vertical speed loop utilizes a compensator to convert vertical speed error into a pitch rate command, while the inner pitch rate controller provides finer pitch adjustments. The auto-throttle loop requires the system to regulate airspeed while maintaining the desired vertical speed and would need an analysis of the cross-coupling effects of the system.

This study utilizes MATLAB software, specifically using System Identification Toolbox [2], Control System Toolbox [3], and Simulink Toolbox [4], to model, design, and simulate the autopilot system. These tools facilitate the identification of system dynamics and the development of control strategies based on theoretical models, which are then validated through time-domain simulations. As discussed in the course materials by Boyang Li [5], MATLAB's System Identification Toolbox is instrumental in deriving accurate models for aircraft dynamics, while the Control System Toolbox aids in designing feedback controllers that meet stability and performance requirements outlined by the scope of this project.

2 Q1 – Vertical Speed Transfer Function

To assess the feasibility of using the aircraft elevator (δ_e) to directly control the vertical speed (v_s), the transfer function $G_{\delta_e}^{v_s}$ was determined. Where $C = [0 \quad -1 \quad 0 \quad V_{trim}]$, the Transfer Function for v_s was found to be (2.1).

$$G_{\delta_e}^{v_s}(s) = \frac{8.57(s - 16.89)(+12.79)(s + 0.04162)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (2.1)$$

The stability of aircraft's control system can be determined by the location of its poles, shown in Figure 1. Poles can be seen on the right half of the s-plane, demonstrating instability.

Further, when analysing the Bode Plot of $G_{\delta_e}^{v_s}(s)$ in Figure 2, a resonant peak at a low frequency can be seen, indicating high sensitivity and possible instability. The phase response can be lagging at low

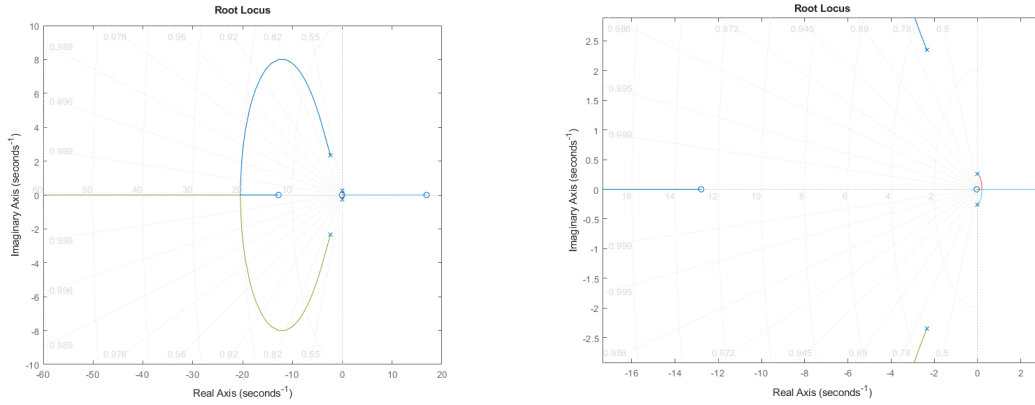


Figure 1: Root Locus of $G_{\delta_e}^{v_s}(s)$

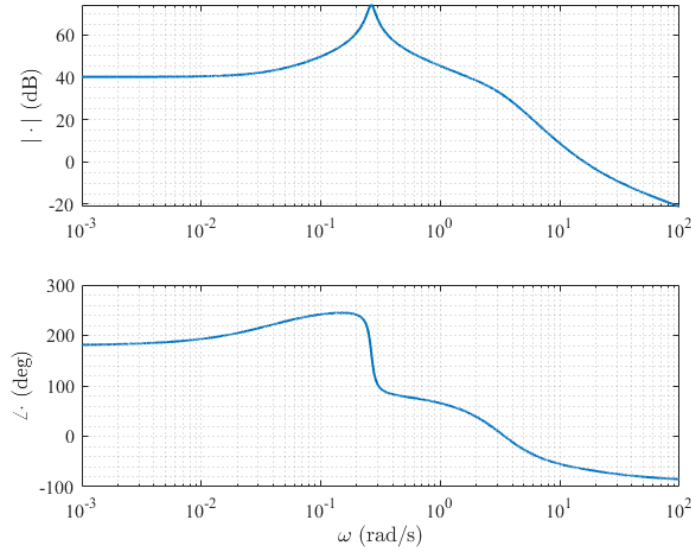


Figure 2: $G_{\delta_e}^{v_s}(s)$ Bode Plot

frequencies. Thus, designing an autopilot using elevator to directly control vertical speed would be challenging due to the indirect relationship between elevator input and vertical speed output.

3 Q2 – Pitch Rate Autopilot

The primary function of the elevator is to control the pitch, which influences the angle of attack, which then in turn affects the lift produced by the wings. When the pitch angle is altered, there is a delayed response as the aircraft's trajectory adjusts to the new wing orientation. This is because the resultant lift force must change sufficiently to alter to the vertical component of the aircraft's velocity.

The Transfer function for elevator deflection to pitch rate q was found to be Equation 3.1, with root

locus of that in Figure 3. The step response of the system was unstable, with 'Inf' overshoot and did not settle. Where $C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$,

$$G_{\delta_e}^q(s) = \frac{-26.397s(s + 1.767)(s + 0.08172)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (3.1)$$

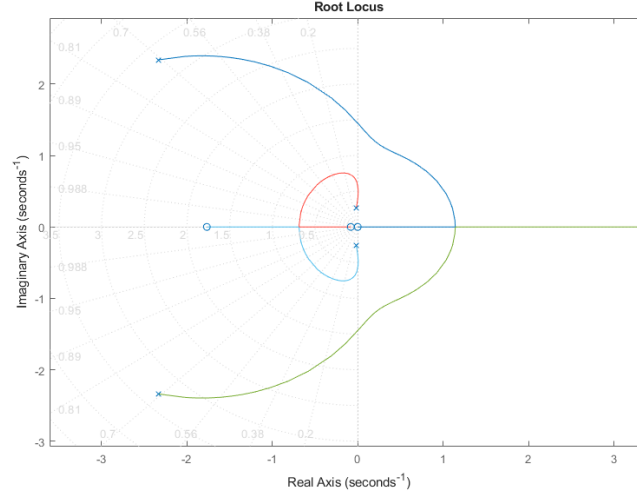


Figure 3: $G_{\delta_e}^q(s)$ Bode Plot

In order to design a controller to satisfy a bandwidth of $\omega_c = 2 - 10 \text{ rad/s}$, noise bandwidth of 20 rad/s, overshoot within 2% with settling time of around 3s, it was determined that that system should have damping ratio (ζ) of 0.78 and Phase Margin of approximately 69.

A proportional integral derivative (PID) controller, with an additional pole, was implemented. The PID parameters were initially estimated and subsequently fine-tuned using the MATLAB 'Control System Designer' tool. Thus resulting in the controller in Equation 3.2.

$$C = \frac{-0.13015(s + 7.692)(s + 0.5263)}{s(s + 1.22e - 5)} \quad (3.2)$$

After applying the PID controller 3.2, the system had settling time of 2.7s and overshoot 8%.

The Bode Plot in Figure 5 shows the system Phase Margin of 64.5 and frequency 5.74 rad/s, close to the desired requirements. While the designed pitch rate autopilot improved the pitch rate control performance, it did not fully meet all the specified criteria. The controller achieved desired performance in terms of settling time and Phase Margin, and effectively attenuates noise above 20 rad/s. However the response has 5% more overshoot than desired.

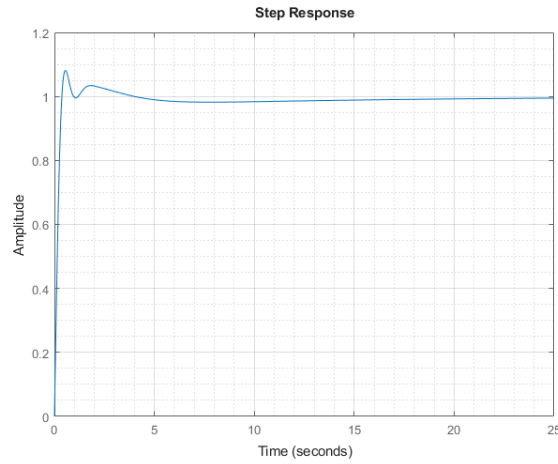


Figure 4: Step response of Controlled System

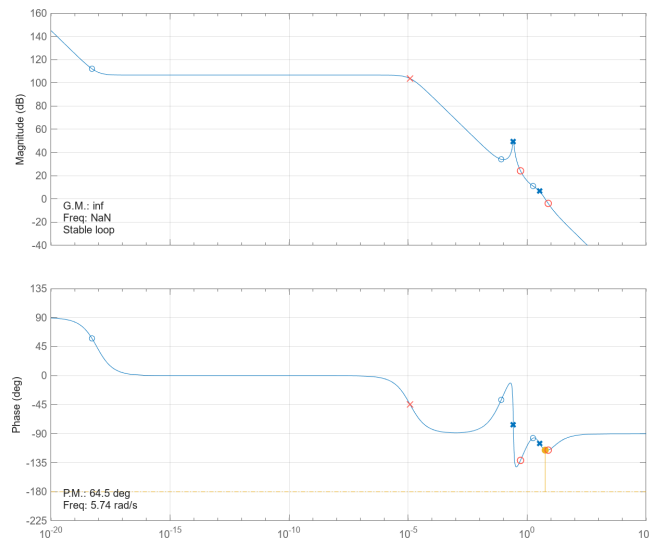


Figure 5: Bode Plot of Controlled System

4 Q3 – Vertical Speed Guidance Loop

The vertical speed guidance loop is designed to control the aircraft's ascent and descent rates by incorporating a pitch rate controller as an inner loop. This loop includes a compensator in the forward leg, which processes the vertical speed error to generate a pitch rate command. The pitch rate command then feeds into the inner loop to modulate the elevator input, allowing the system to adjust vertical speed in response to deviations from the desired reference.

4.1 Loop Topology

The complete loop topology of the vertical speed guidance loop is shown in Figure 6. The compensator transforms the vertical speed error ($vs_{ref} - vs$) into a pitch rate command. This command is then processed within the inner loop section ($de2q$), where the transfer functions G_1 and G_2 are used in the pitch rate controller to compute the required elevator deflection (δ_e) to achieve the target pitch

The vertical speed response, shown in Figure 2, demonstrates the system's performance in tracking a 500 ft/min (approximately 2.54 m/s) vertical speed command step input. The response shows a rapid increase in vertical speed, reaching close to the desired 2.5 m/s target within the first 2 seconds. There is a slight overshoot beyond the target speed, followed by a gradual settling towards the reference value, achieving stability within approximately 10 seconds. This response indicates that the system meets the rise time and settling time requirements, effectively balancing speed of response with controlled damping. The smooth convergence to the command confirms the system's capability to achieve accurate vertical speed tracking.

4.3 Pitch Rate Response

The pitch rate response, shown in Figure 8, reflects the performance of the inner pitch rate controller.

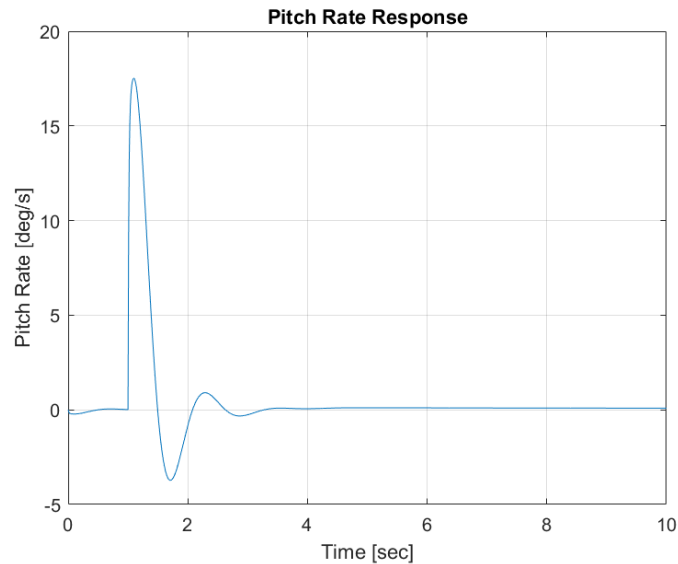


Figure 8: Pitch Rate Response

The response shows a peak reaching 16 deg/s , followed by a quick correction with minor oscillations before settling to zero. This peak and correction indicate the controller's rapid adjustment to the pitch rate required to meet the vertical speed command of 500 ft/min (2.54 m/s), as outlined in Section 4.2. These damped oscillations demonstrate the pitch rate controller's stability and responsiveness.

4.4 Elevator Activity

Figure 9 displays the control activity of the elevator in response to the inputted vertical speed command.

The elevator initially deflects sharply to approximately -2.3 degrees as the controller adjusts to achieve vertical speed command, followed by the elevator deflection stabilizing back to zero, demonstrating its minimal effort required once the target vertical speed is reached.

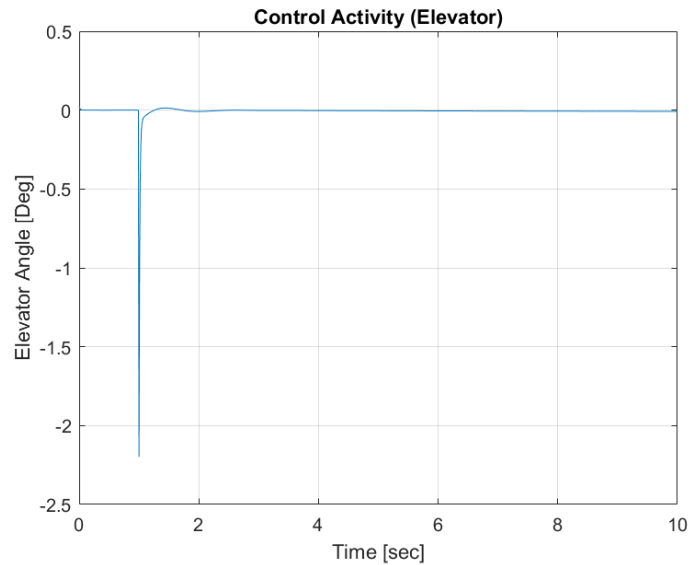


Figure 9: Elevator Control Activity

5 Q4 – Design Auto-Throttle Control System

In this section an auto-throttle design was attempted, wherein errors in airspeed would determine throttle command.

5.1 MIMO Loop Topology

The auto-throttle system was to be achieved with the MIMO loop topology shown in Figure 10.

5.2 Guidance Loop Topology Specifications

This loop was designed to the same specifications as the vertical speed guidance loop, with a vertical speed step input of 500 ft/min. The other constraints could not be implemented however due to difficulties in controlling the system. In addition to the vertical speed input there was also an input for u (airspeed).

5.3 Primary Control Effects

The primary control effects, namely the transfer function of throttle to airspeed ($(dt2u)$), and elevator to vertical speed ($de2vs$), are shown in Figures 11 and 12.

These effects tend to follow a more sinusoidal pattern, reaching similar magnitudes over the simulation period. This is quite different from the responses found from solely the vertical speed control loop (in 4.1). The throttle is more effective in making airspeed changes because of its direct affect on this output, coupled with the effects from the elevator control, although because of the increased complexity of the system it is more difficult to tune, and does not reach a stable point (like the outputs of the vertical speed control loop).

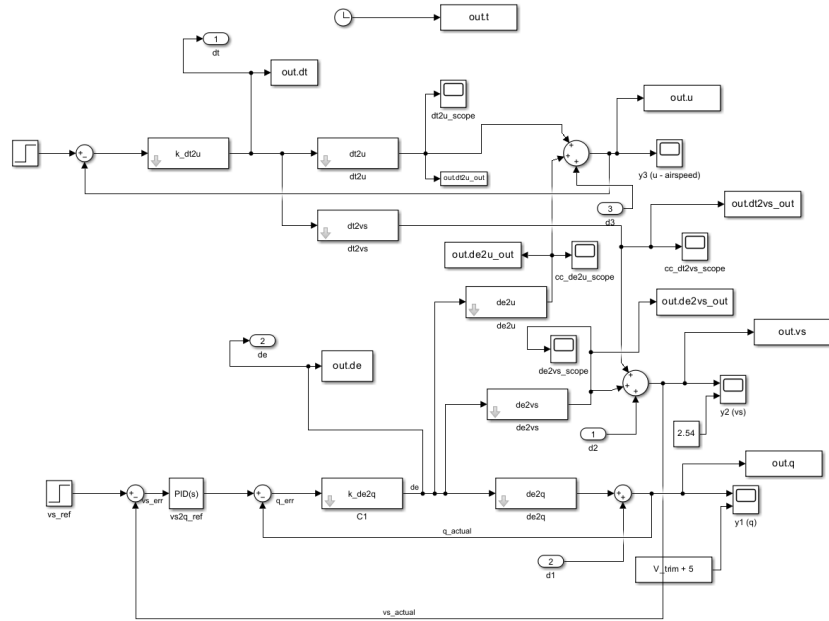


Figure 10: MIMO Simulink Loop for Auto-Throttle Design

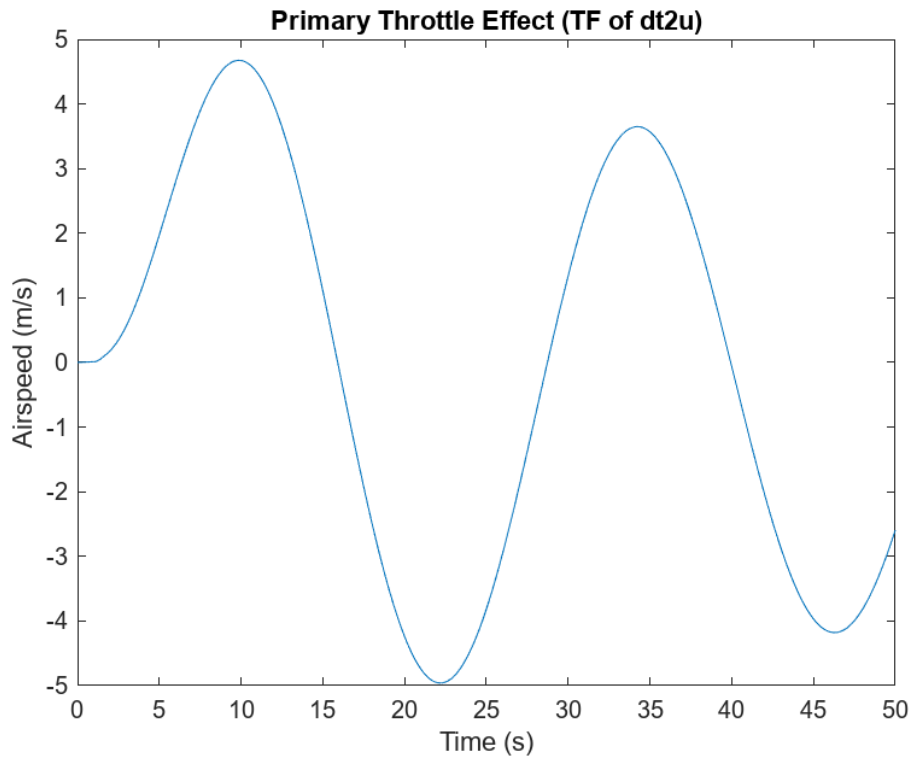


Figure 11: Primary Control Effect: Throttle to Airspeed

The overall output of vertical speed and pitch rate (from both elevator and throttle control) however show a more similar trend to the vertical speed control loop, shown in Figures 13 and 14, albeit

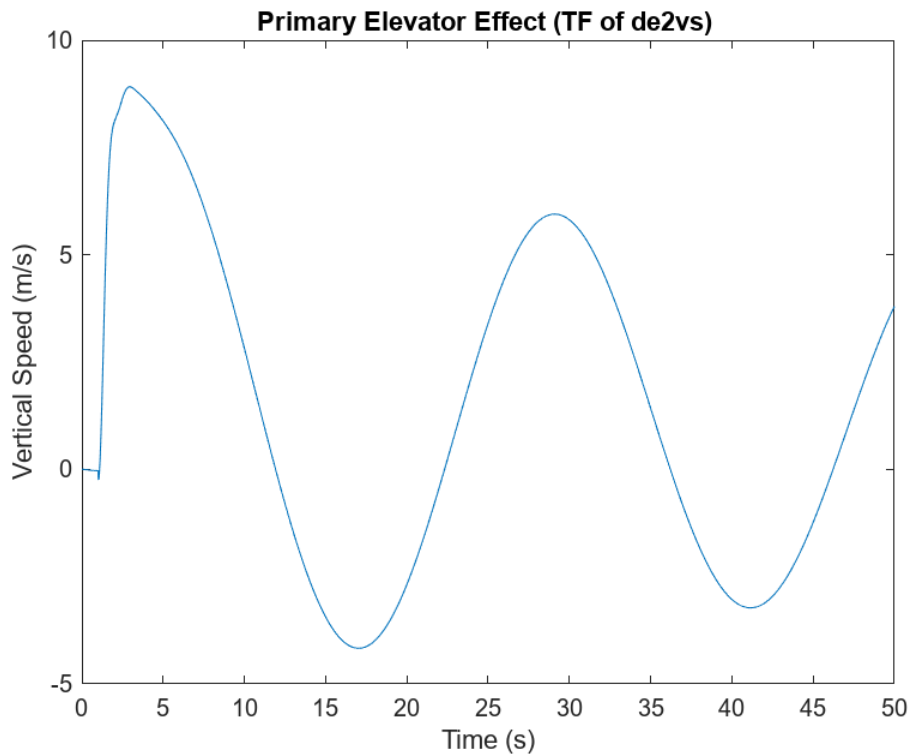


Figure 12: Primary Control Effect: Elevator to Vertical Speed

with a longer settling time (i.e., not being controlled in the simulation time). Airspeed will not be compared as it is not explored in the vertical speed control loop, however it is included for completeness (15).

5.4 Cross-coupling Control Effects

The cross-coupling control effects, namely the transfer function of throttle to vertical speed ($(dt2vs)$), and elevator to airspeed ($de2u$), are shown in 16 and 17.

Again, compared to only the vertical speed control loop, these responses are sinusoidal, and do not stabilise in the timeframe - showing the complexities of cross looping.

5.5 Elevator and Throttle Response

The control activities of the elevator and throttle in response to the inputted vertical speed and airspeed commands are shown in Figures 18 and 19. The throttle follows a reasonable trend, appearing to begin stabilising at 50 seconds in response to the airspeed step input (although not having a stabilising effect on airspeed as seen in Figure 11). The elevator seems to be more erratic, with a large initial deflection, and then no change for the simulation period. This may again be due to the cross coupling effects, and the increased complexity of the system.

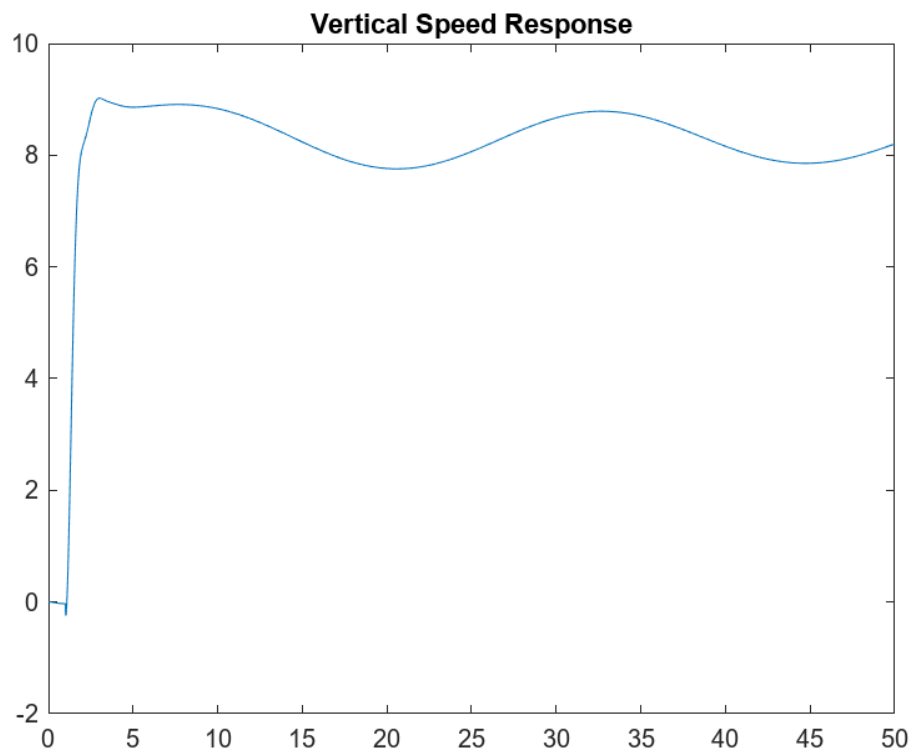


Figure 13: Vertical Speed Response (Auto Throttle System)

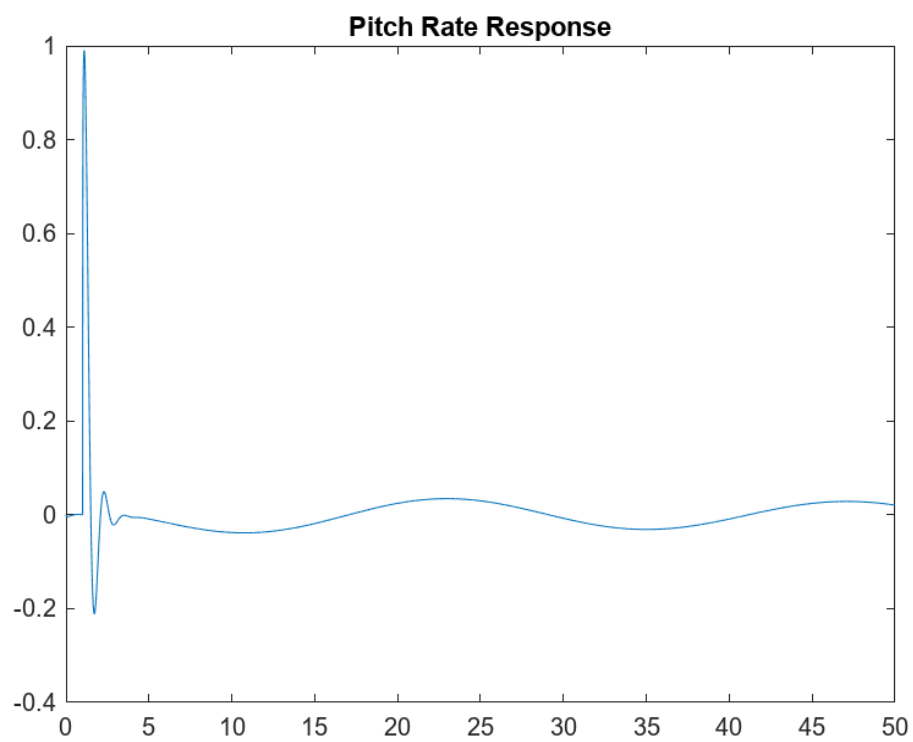


Figure 14: Pitch Rate Response (Auto Throttle System)

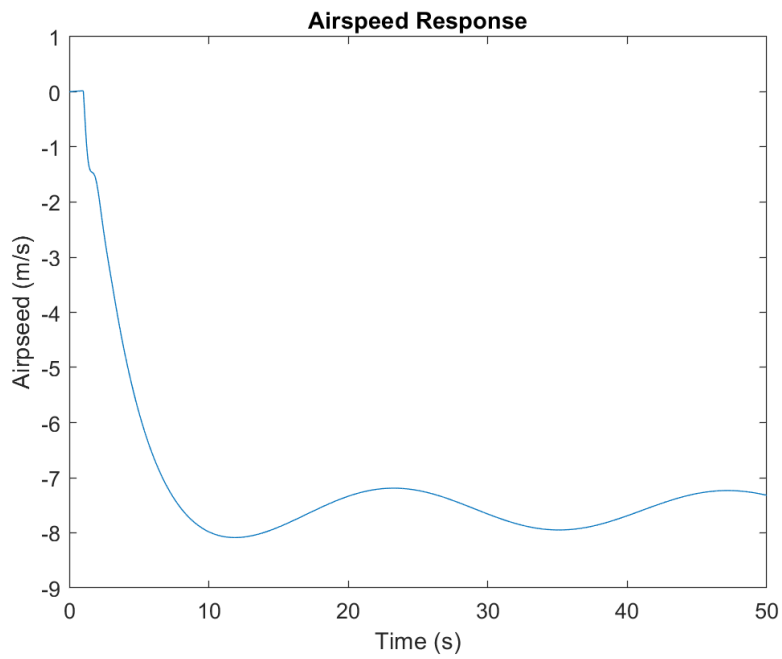


Figure 15: Airspeed Response (Auto Throttle System)

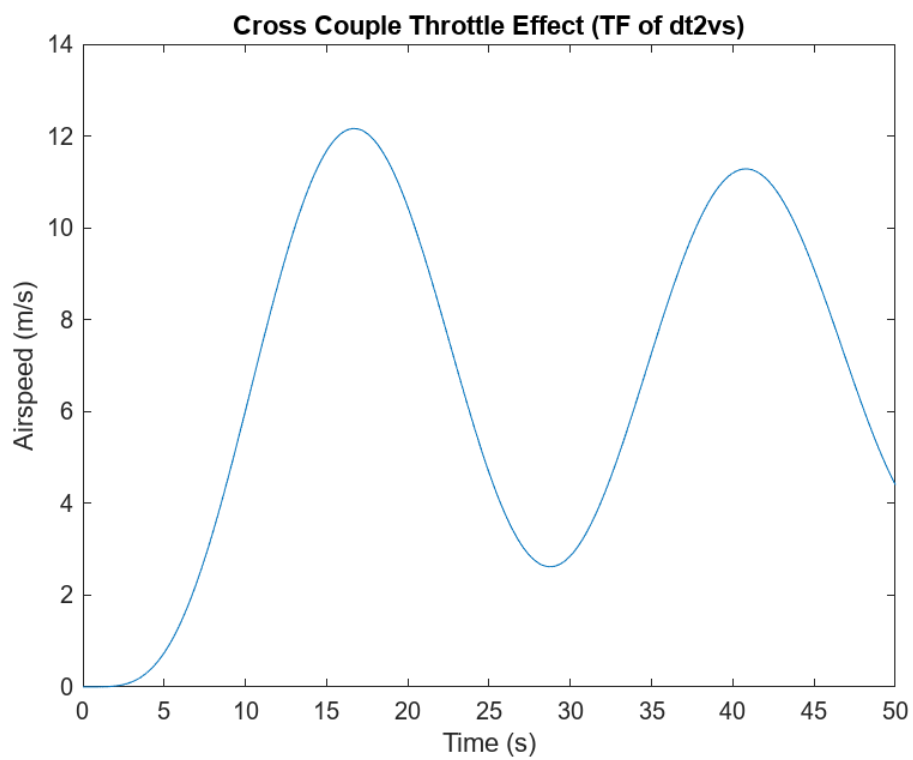


Figure 16: Cross Couple Response (Throttle to Vertical Speed)

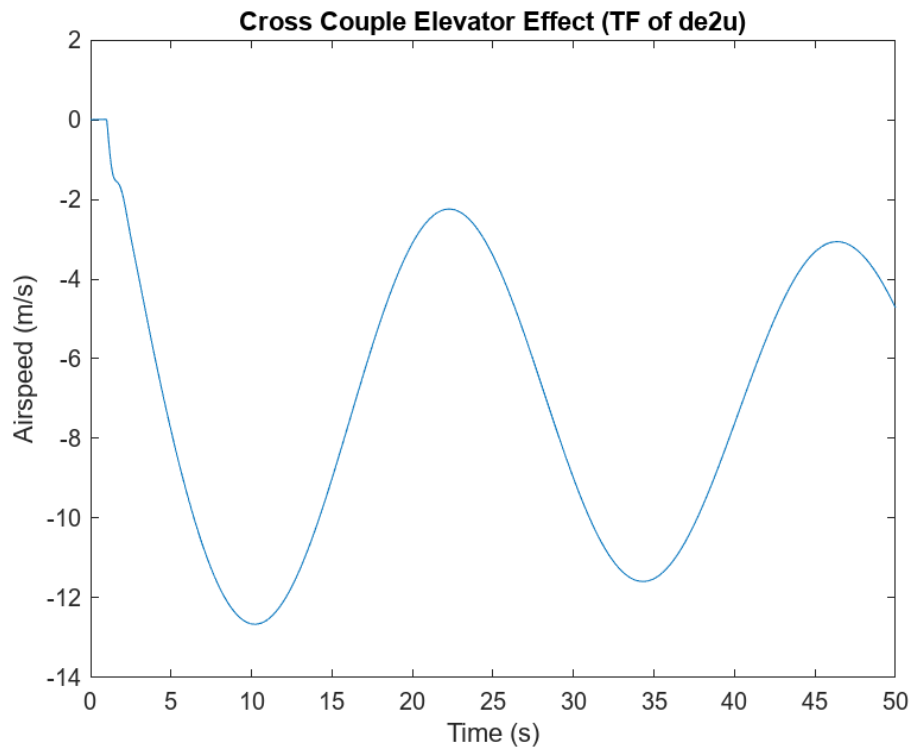


Figure 17: Cross Couple Response (Elevator to Airspeed)

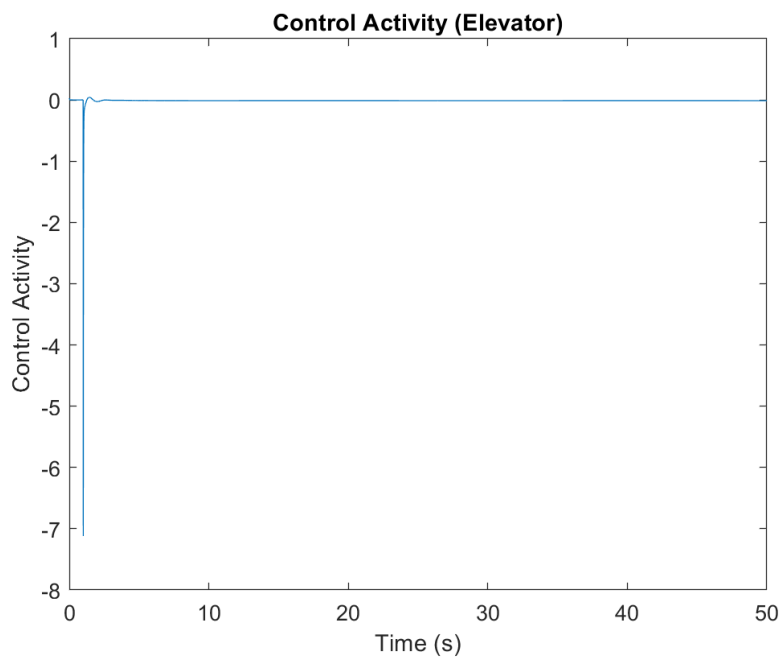


Figure 18: Elevator Control Activity (Auto Throttle System)

5.6 Auto-Throttle Effectiveness

The effectiveness of the auto throttle system without the vertical speed controller was explored, both with and without the pitch rate controller. It was discovered that the vertical speed controller is

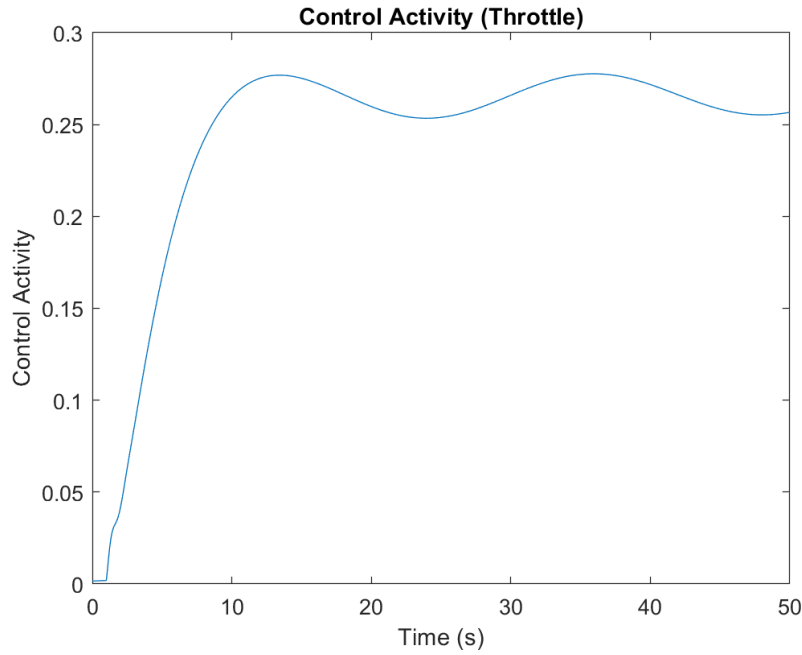


Figure 19: Throttle Control Activity (Auto Throttle System)

necessary for non-erratic behaviour of airspeed, as seen in Figure 20. Furthermore, deactivating vertical speed along with the pitch rate controller made the system unable to be simulated (perhaps due to being too unstable). The error is shown in Figure 21.

6 Final Design

6.1 Q5 – Implementation of the control and guidance laws

A six degree of freedom (DOF) simulation includes all translational and rotational motions, capturing complex interactions and nonlinearities, unlike the transfer function models which simplify the behaviour of the system around a specific point.

In Figure 22 and Figure 23, the MIMO controller was represented as a state-space system.

The linear results of Figure 22 are as expected, similar to the results of the SISO model in Section 5. The coupling effects of the system are highlighted when analysing the individual state responses in state-space form. In response to the commanded 500 ft/min vertical speed control input, the system stabilises after approximately 7 seconds.

The nonlinear responses in Figure 23 appear more damped and exhibit less sensitivity to inputs, suggesting that the control law might be effectively compensating for changes. However, it is more plausible that this behavior is due to an incorrect implementation of the state-space model of the system. The variables u , w , q , and θ all show responses with reduced amplitude. While these differences could be attributed to nonlinear dynamics not accounted for in the linear model, they are more likely the result of inaccuracies in the model implementation.

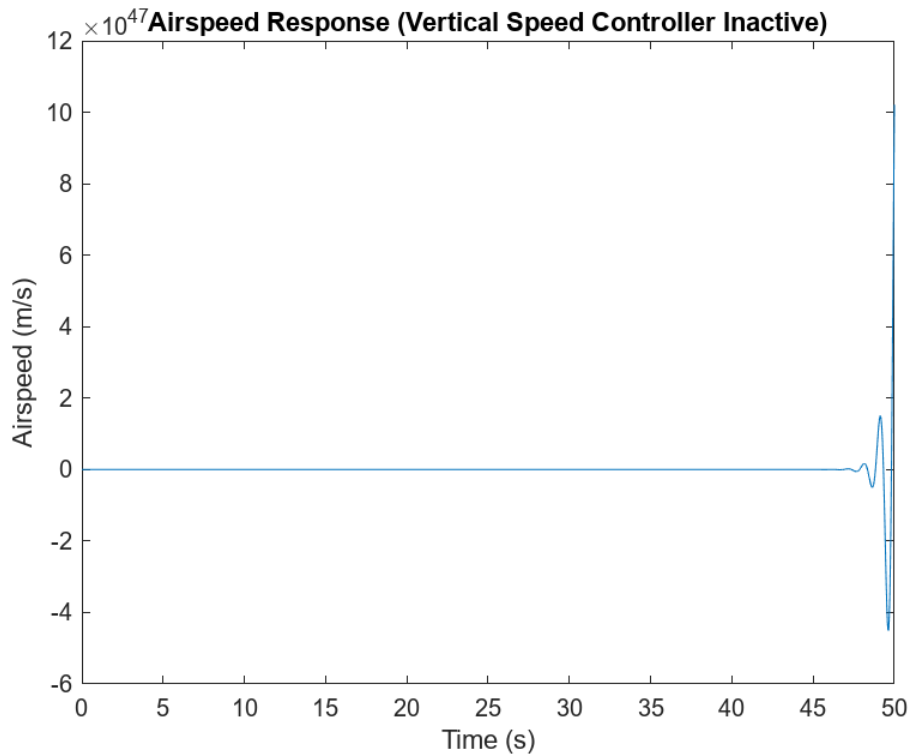


Figure 20: Airspeed Response with Vertical Speed Controller Deactivated

Continuous state '1' of 'simulink_of_20/debu/Intermal' at time 33.96 is not finite. Simulink will terminate the simulation. There may be a singularity in the solution. Check the model 'simulink_of_20' for errors. Try reducing the fixed step size or tighten the error tolerances for variable-step solvers.

Figure 21: Error Due to Vertical Speed Controller and Pitch Rate Controller Being Deactivated

6.2 Q6 – Gust Effectiveness

By simulating gust effects in both linear and nonlinear models, this analysis evaluates the control system's ability to counteract these disturbances and restore the aircraft to its desired flight state. The comparison between linear and nonlinear responses provides insights into the robustness of the control system across different modeling complexities.

In the linear model simulation (Figure 24), the closed-loop control system shows a strong capability to counteract gust disturbances. The airspeed (u) initially experiences a minor drop but quickly stabilizes as the control system compensates for the disturbance. Similarly the vertical speed (w) and the pitch angle (θ) responses show minor deviations. Vertical velocity initially peaks but settles back to the desired value within a few seconds, while the pitch angle undergoes a small fluctuation before returning to a steady-state position. The linear model's behavior suggests that the control system, is able to resist the external gust effects and is able to maintain the desired flight parameters.

The nonlinear model simulation (Figure 25) provides a more realistic assessment of the system's gust rejection performance by assessing the real-world aircraft dynamics. In response to the gust disturbance, the closed-loop system in the nonlinear model shows slightly larger transient deviations compared to the linear model. Airspeed (u) initially drops more significantly, and vertical velocity (w) experiences a sharper peak. The pitch angle (θ) also shows a larger initial fluctuation before stabilizing. Despite these increased responses, the control system manages to counteract the disturbance, returning

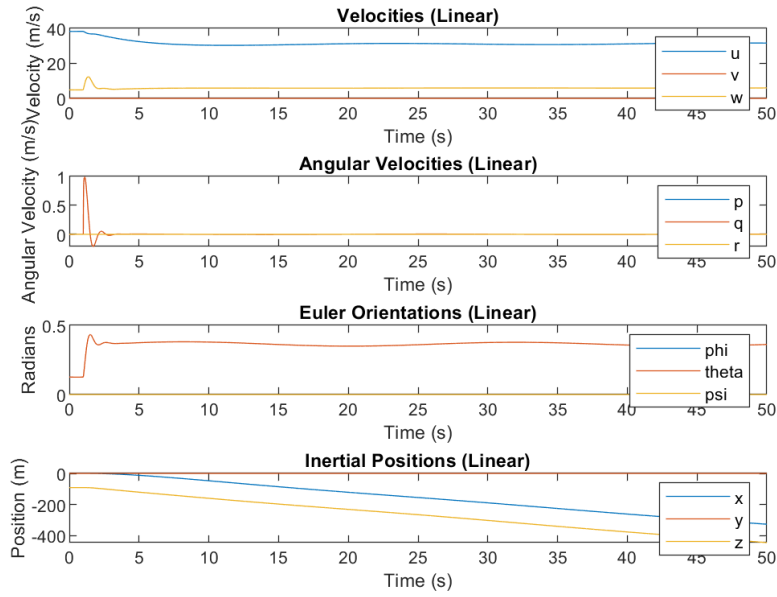


Figure 22: 6 DOF State-Space System Linear Response

the variables to near-steady-state conditions after a short duration, meaning that the nonlinear model is robust even when under realistic dynamics conditions even though it experiences a larger overshoot and increased settling time.

In both cases, the control system demonstrates its capacity to reduce gust-induced disturbances, validating its robustness in both simplified and realistic dynamic scenarios, however the nonlinear model's responses indicate that the system needs to accommodate larger deviations and longer recovery times.

6.3 Q7 – Commanded Large Vertical Speed

Commanding a large vertical speed input in the aircraft would introduce a number of challenges as the aircraft's system control is not very robust. Such commands can drastically increase aerodynamic loads, exacerbating existing instabilities and leading to dangerous oscillations or aircraft stall.

To address these challenges, a more advanced control strategy could be implemented. Implementing an adaptive control can further enhance the system by dynamically adjusting control gains based on current flight conditions and command magnitudes. This control strategy could replace or augment the existing PID controllers in the MIMO 6 DOF simulation of Section 5. A predictive control, such as Model Predictive Control (MPC), predicts future states to adjust control inputs preemptively, enhancing the system's ability to maintain stability by optimising tracking performance and control effort.

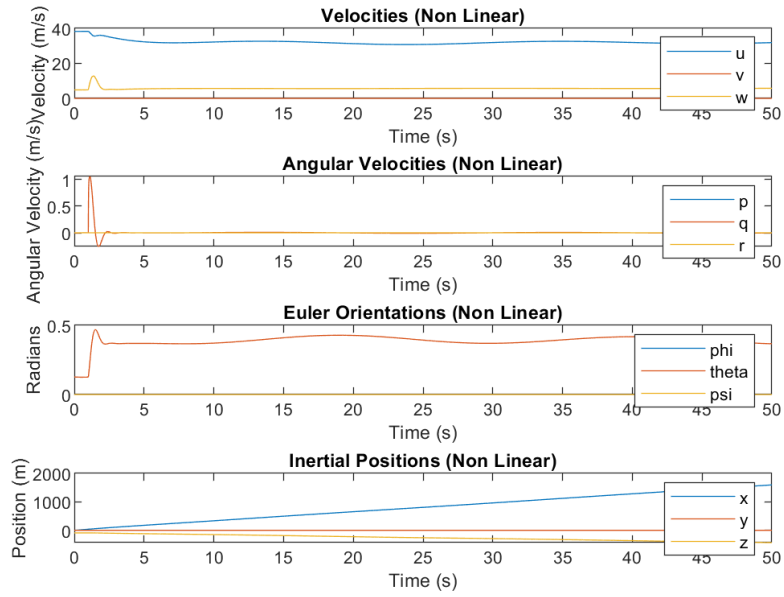


Figure 23: 6 DOF State-Space System Nonlinear Response

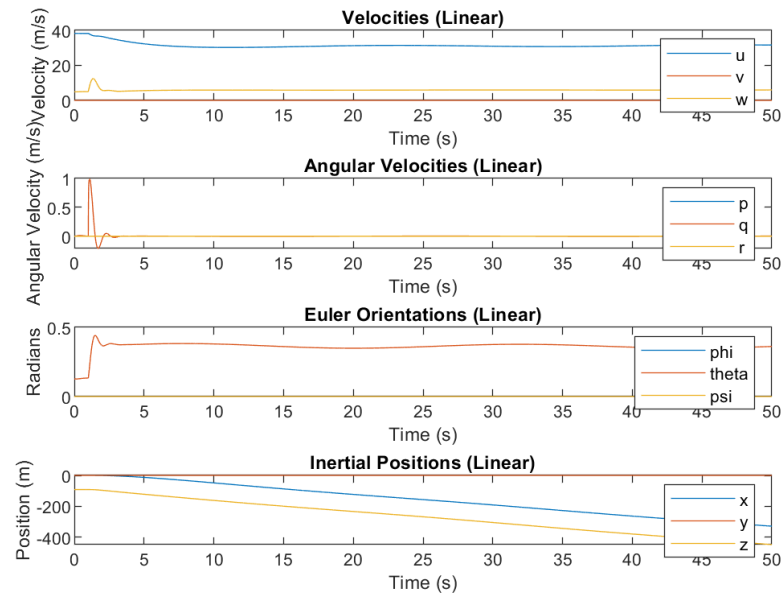


Figure 24: 6 DOF State-Space System Linear Response to Gust

7 Conclusion

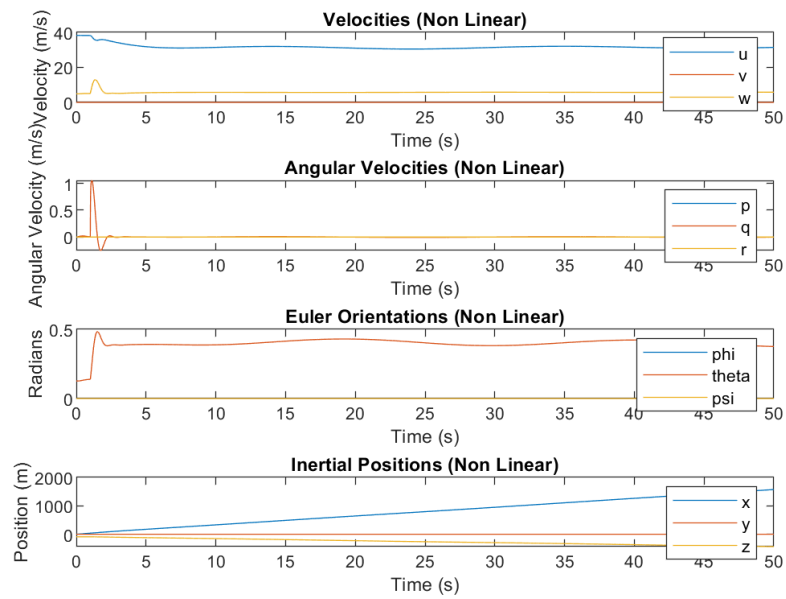


Figure 25: 6 DOF State-Space System Nonlinear Response to Gust

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